Chapter 12. Gasping for Oxygen

The cycling of oxygen between the breathable oxygen molecule O_2, and carbon dioxide CO_2, and also (more slowly) other forms, is called the Earth's "oxygen cycle." Google it. The cycling of carbon between CO_2 in the atmosphere, plants (dead and alive), and also (much more slowly) other forms such as coal, oil, and kerogen, is called the "carbon cycle." Google it, too. Obviously these cycles are coupled; they influence each other. They are the foundation for Chapter 13.

Chapter 13. Interstellar Travel

Exoplanets (planets beyond our solar system) are being discovered at a furious pace. Nearly complete catalogs, updated daily, are at http://exoplanet.eu and http://exoplanets.org. A catalog of exoplanets that could be habitable is at http://phl.upr.edu/hec. For the human side and history of the search for exoplanets and life beyond the solar system, see *Mirror Earth: The Search for Our Planet's Twin* (Lemonick 2012) and *Five Billion Years of Solitude: The Search for Life Among the Stars* (Billings 2013); for technical and scientific details, see *The Exoplanet Handbook* (Perryman 2011). *Confessions of an Alien Hunter: A Scientist's Search for Extraterrestrial Intelligence* (Shostak 2009) is an excellent description of the search for extraterrestrial intelligence (SETI) via radio signals from beyond Earth and by other methods.

For information about technologies that we humans could pursue in our quest for interstellar travel, I suggest http://en.wikipedia.org/wiki/Interstellar_travel and http://fourthmillenniumfoundation.org. The astronaut Mae Jemmison is spearheading a quest to send humans beyond the solar system in the next century; see http://100yss.org. A lot of nonsense is written about interstellar travel via warp drives and wormholes. The technology of